

MODUL VISUALISASI DATA

Bar Chart & Line Chart (Tableau)

A. Capaian Pembelajaran

1. Memahami konsep dasar visualisasi data.
2. Mengenal antarmuka dasar pada Tableau.
3. Memahami perbedaan dan penggunaan **bar chart** dan **line chart**.
4. Mampu membuat berbagai jenis bar chart dan line chart.
5. Mampu melakukan customisasi visualisasi agar menarik dan mudah dibaca.
6. Mampu membuat **combo chart (bar + line)** sederhana.

B. Materi Praktikum

1. **Pengenalan Tableau & Dataset**
 - a. Interface dan data (dimension & measure)
2. **Bar Chart**
 - a. Simple, stacked, grouped, horizontal, 100%
 - b. Sorting, warna, label
3. **Line Chart**
 - a. Simple, multiple, marker, area, dual line
 - b. Format waktu, warna, trend
4. **Combo Chart (Bar + Line)**
 - a. Dual axis (Sales & Profit)
5. **Latihan**
 - a. Bar chart
 - b. Line chart
 - c. Combo chart

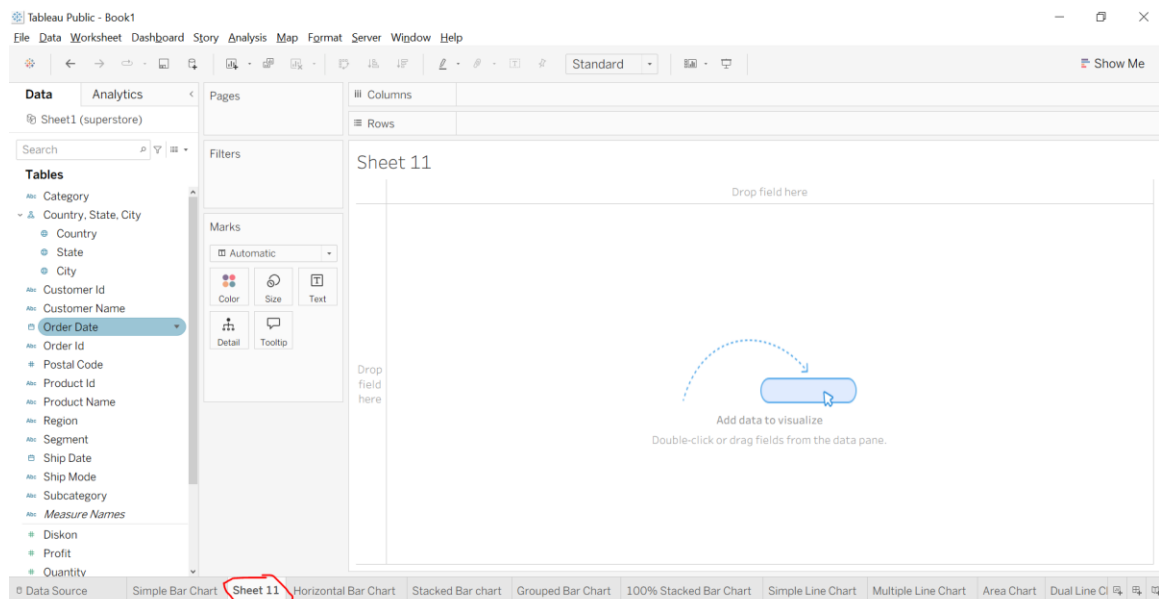
1. Bar Chart

Bar chart digunakan untuk membandingkan nilai antar kategori.

Jenis-jenis Bar Chart:

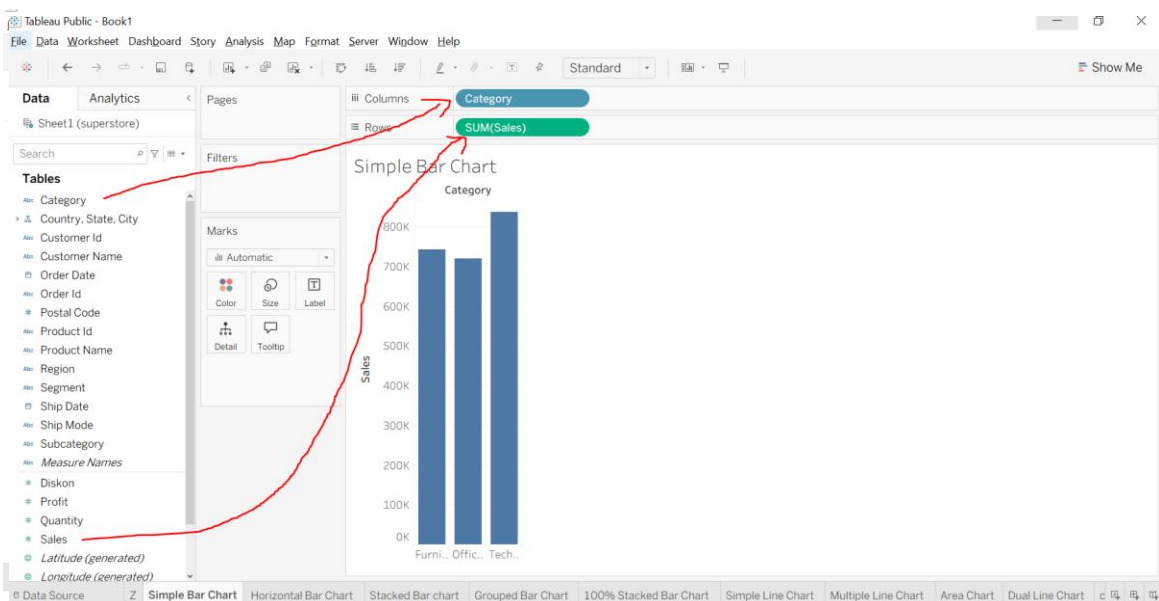
- **Simple Bar Chart**

Ganti Nama sheet → Simple Bar Chart.

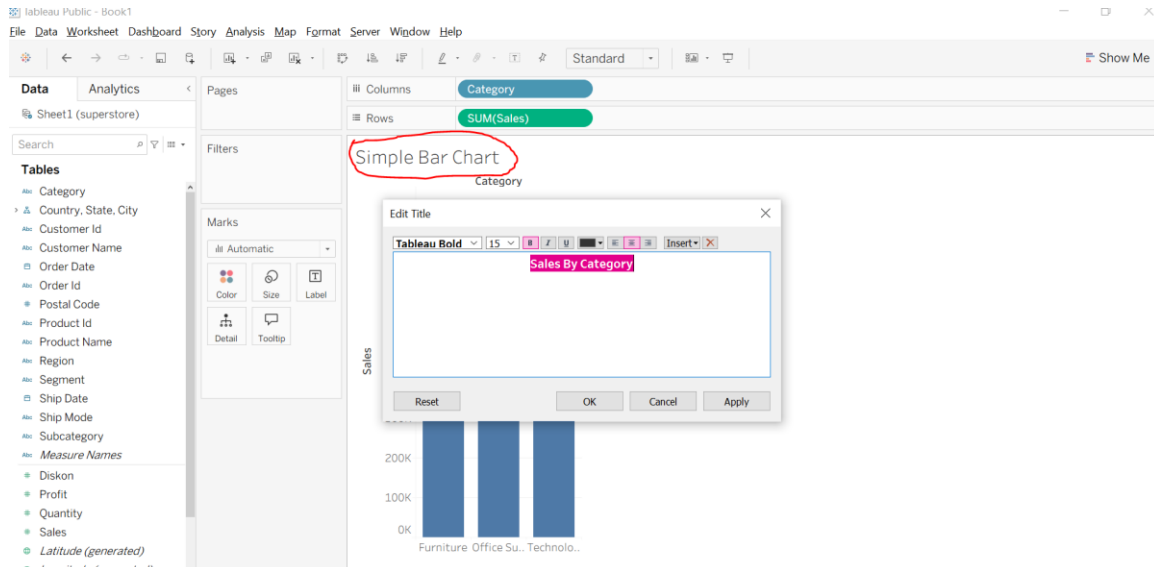


Drag & Drop Kolom Category → Columns

Drag & Drop Kolom Sales → Rows

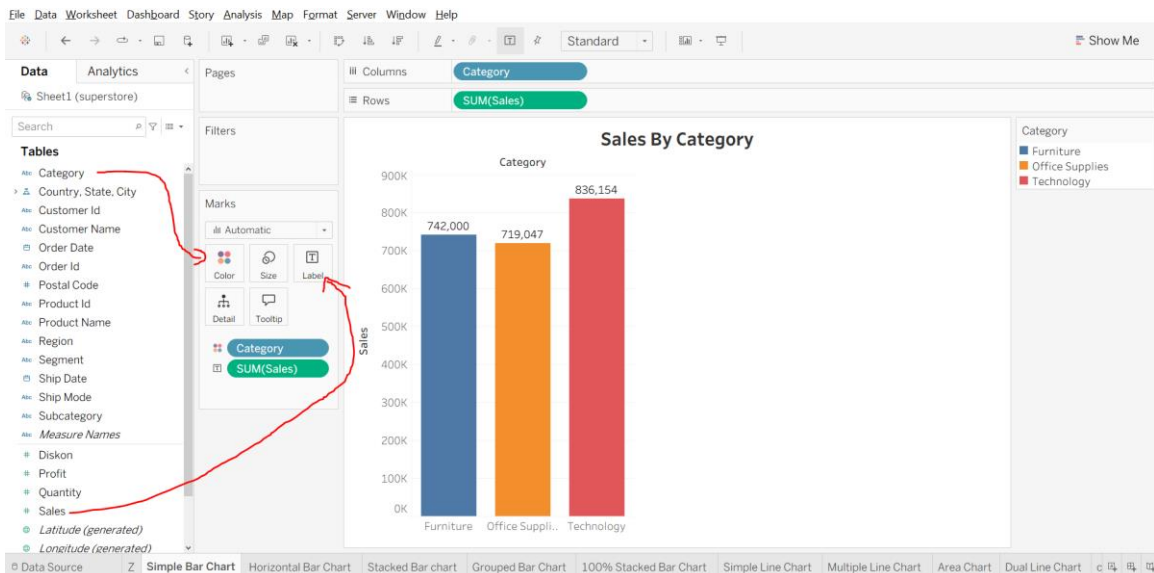


Ganti Judul nya:
Simple Bar Chart → Sales By Category



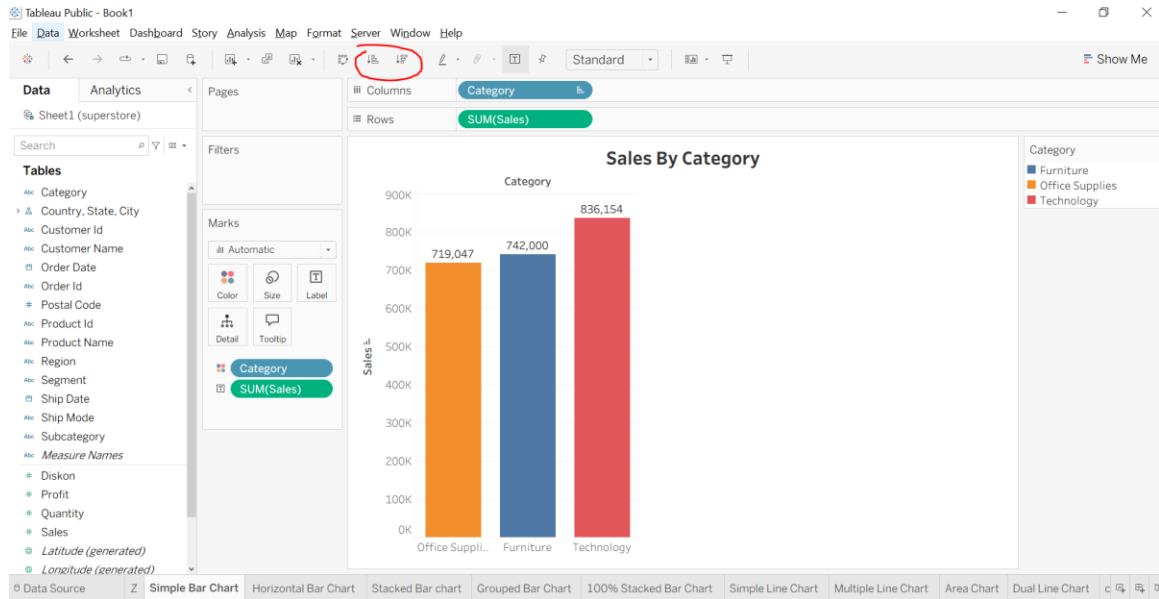
Drag & Drop kembali kolom Category dan Sales pada bagian Marks

Category → Colors
Sales → Label



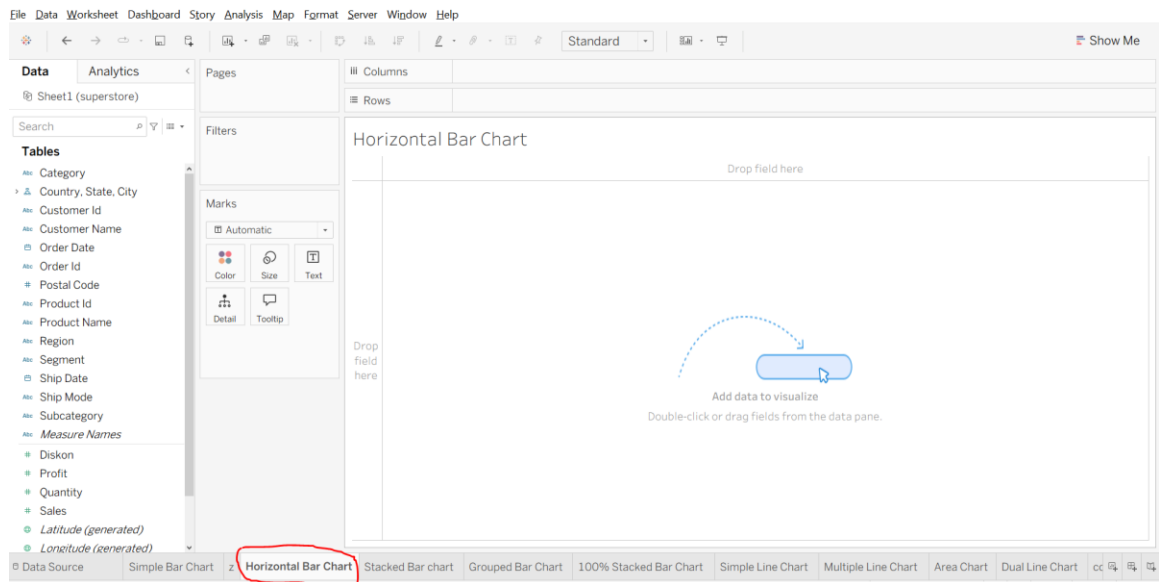
Sehingga Outputnya Terklasifikasikan berdasarkan warna category serta ada label di bagian atas bar nya.

Lalu Sorting data tersebut Descending atau Ascending



- **Horizontal Bar Chart**

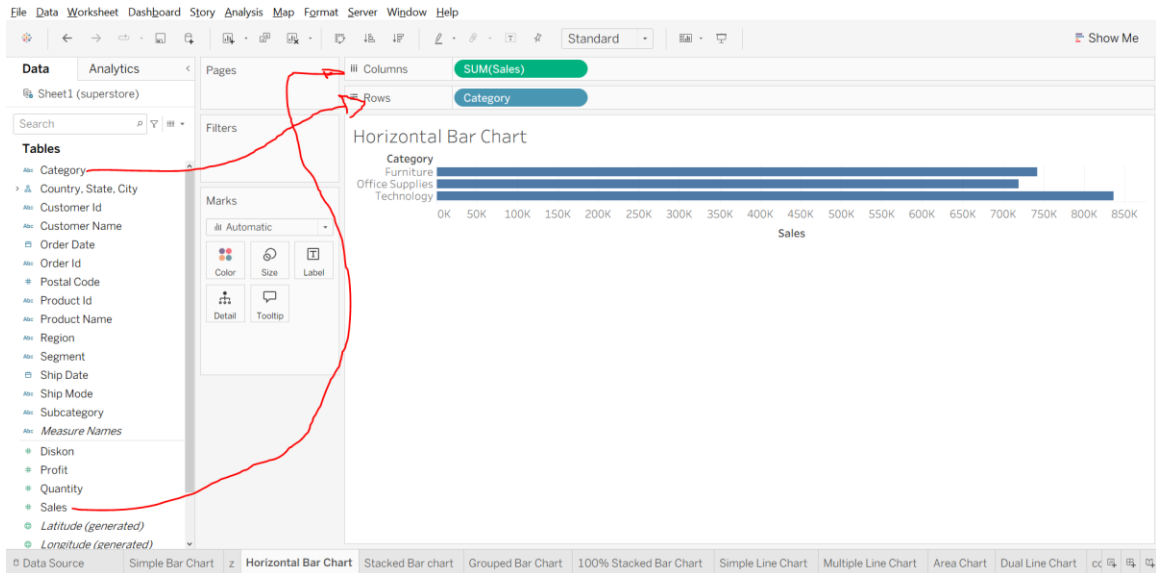
Buat Sheet Baru Kembali dan ganti nama worksheet nya menjadi Horizontal Bar Chart.



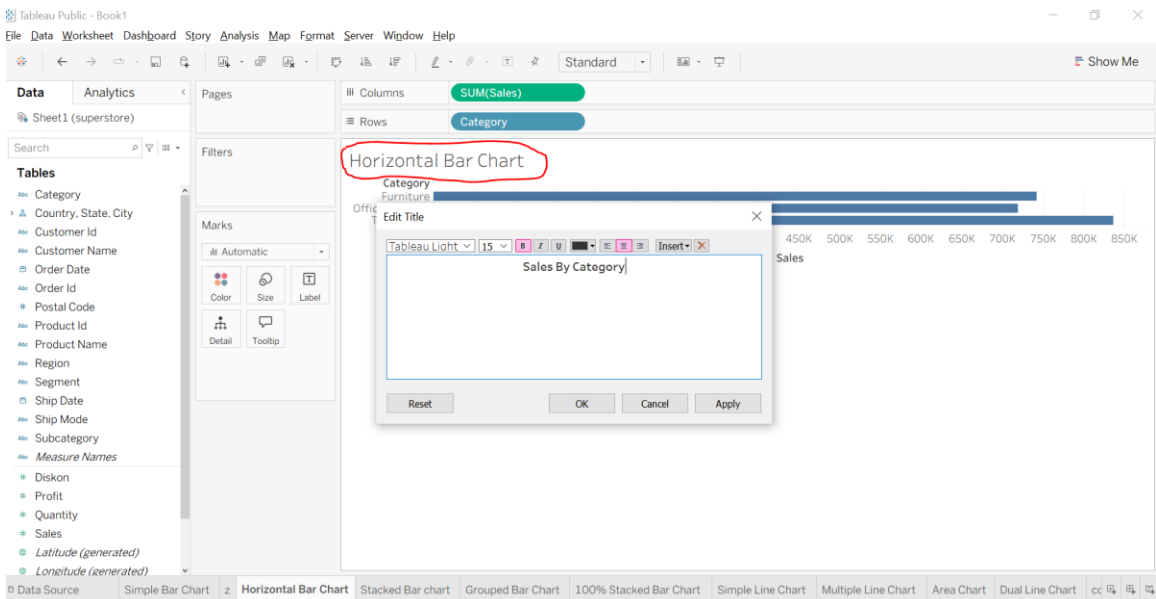
Drag Kolom:

Category → Rows

Sales → Columns



Ganti Title Bar Chart nya → Sales By Category

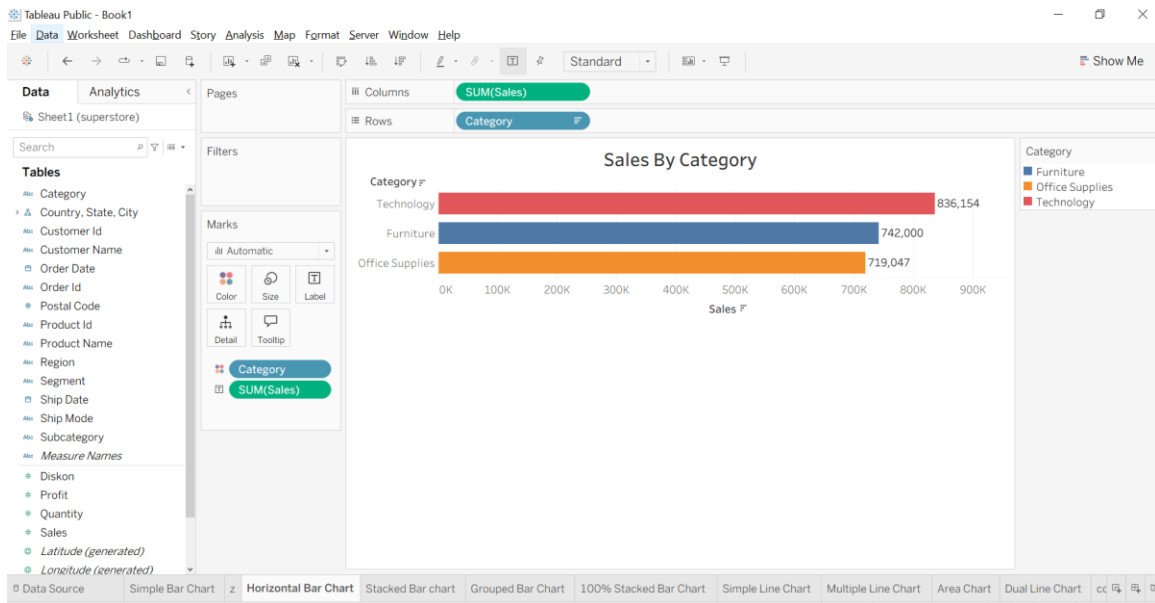


Di bagian Marks, Drag kolom:

Category → Color

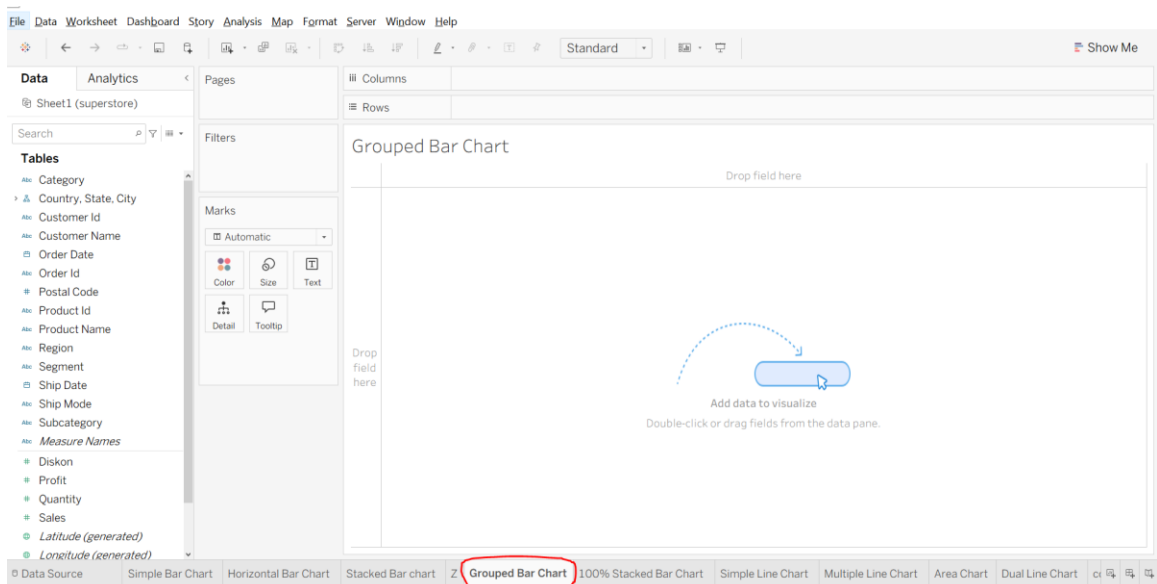
Sales → Label

Lalu sorting/urutkan Dari Data yang terbesar ke Terkecil

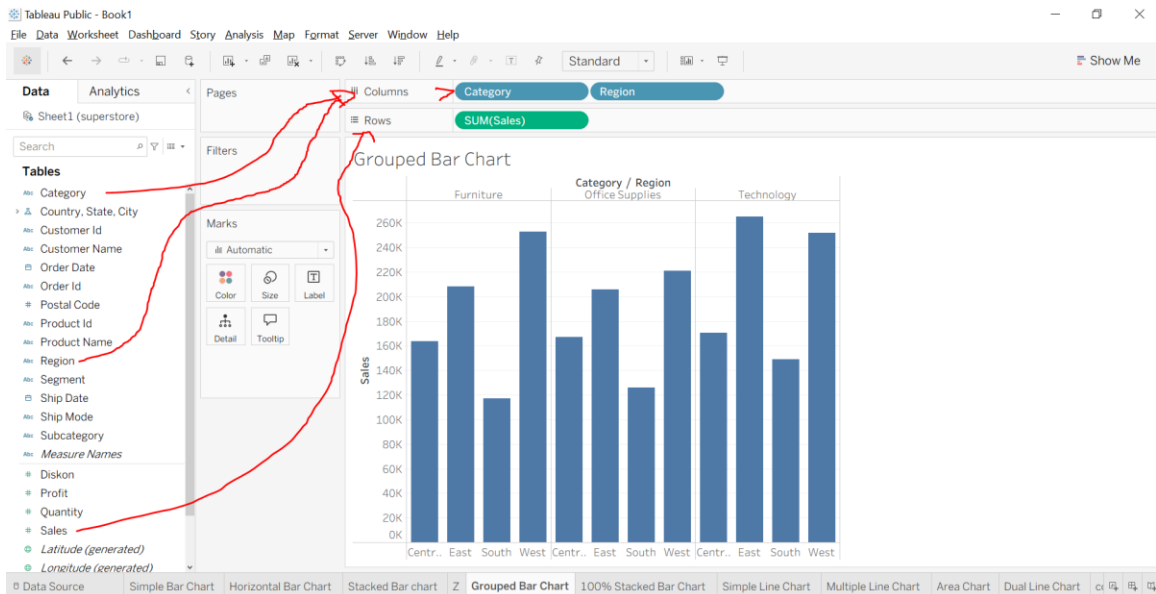


- **Grouped Bar Chart**

Buat Worksheet Baru lagi dan ganti nama sheet nya menjadi Grouped Bar Chart

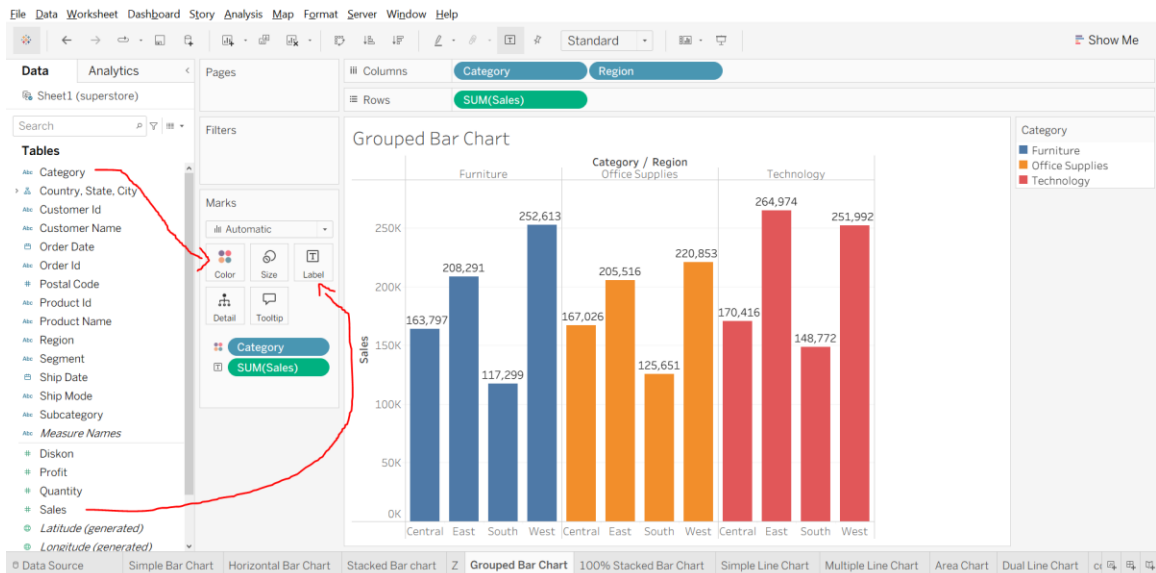


Drag and drop kolom:
 Category → Columns,
 Region → Columns,
 Sales → Rows

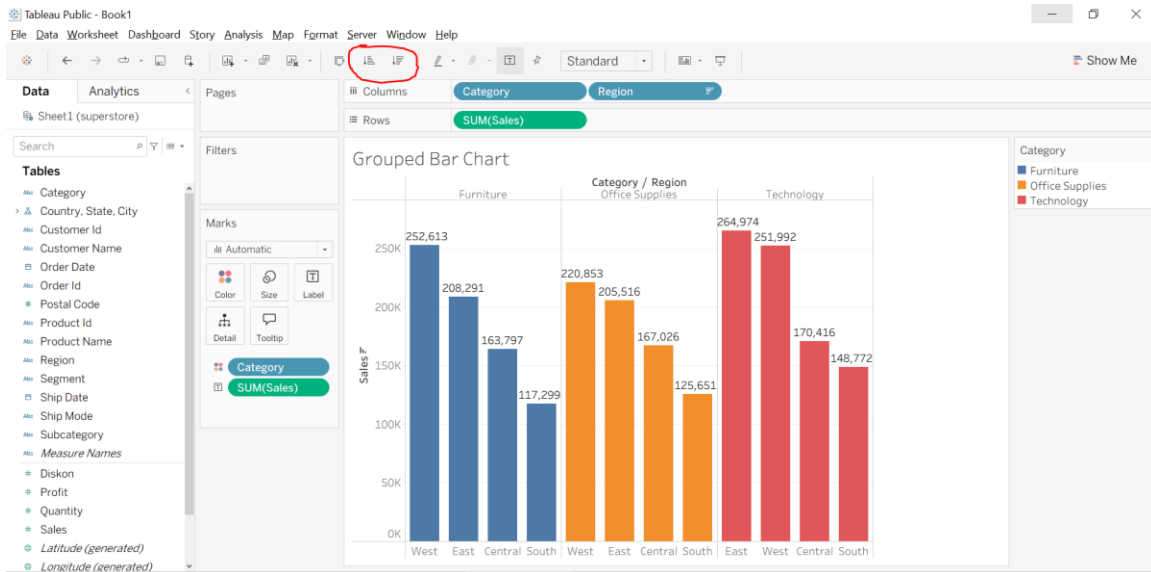


Drag & Drop kolom:

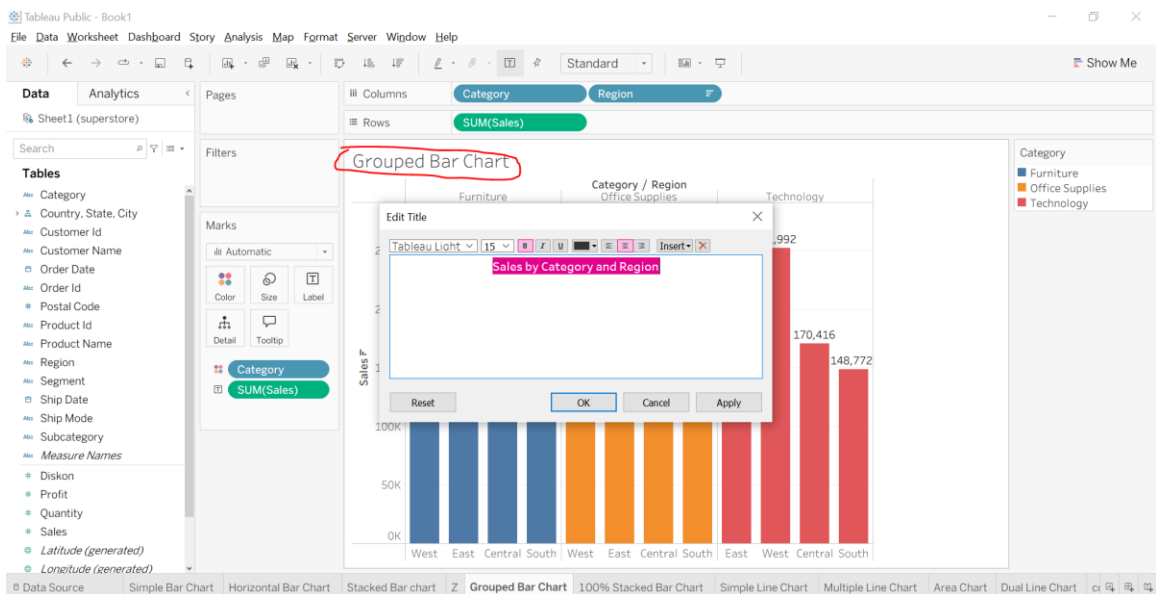
Category → Colors,
 Sales → Label



Lalu Urutkan Secara Descending.

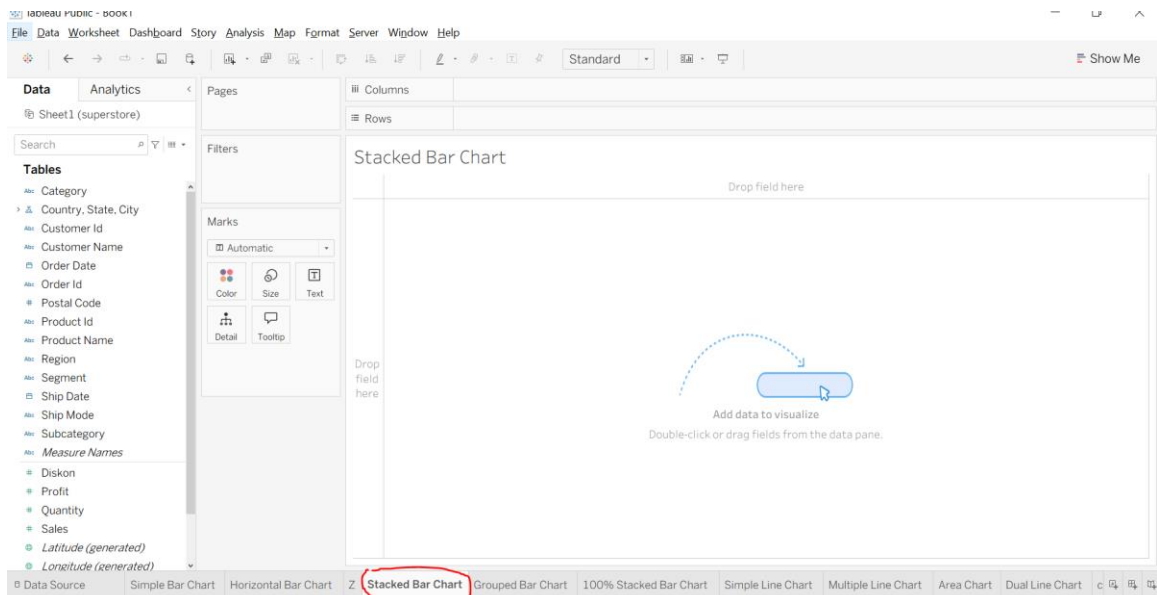


Ubah Title Grouped Bar Chart → Sales By Category and Region



- Stacked Bar Chart

Buat Woorksheet baru Kembali dan Rename Menjadi Stacked Bar chart



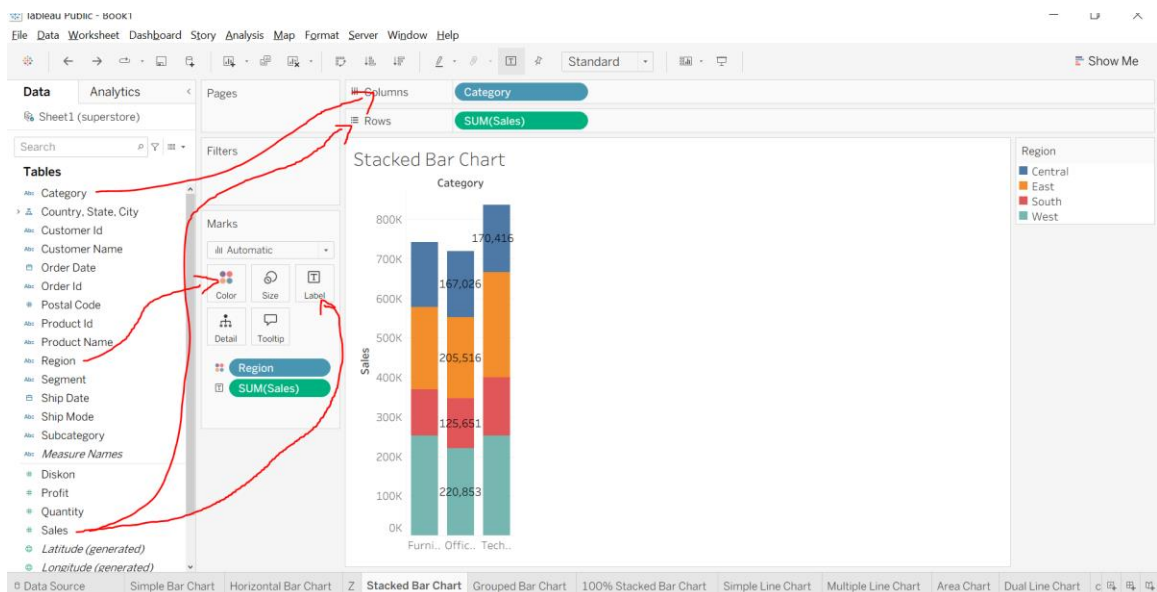
Drag & Drop Kolom:

Category → Columns,

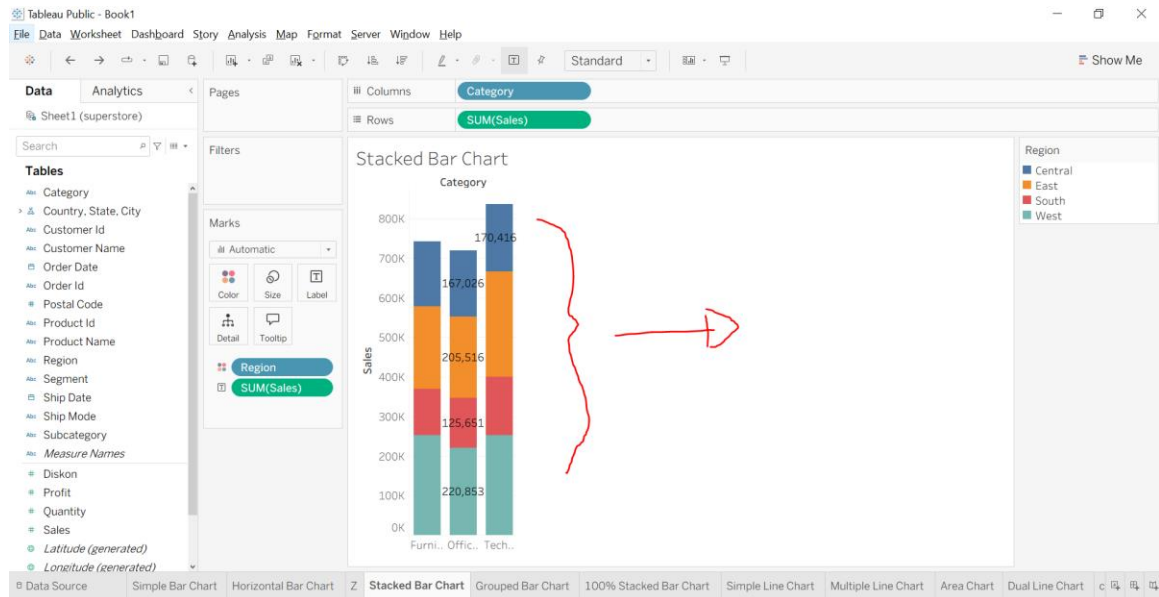
Region → Color (Dibagian Marks),

Sales → Rows,

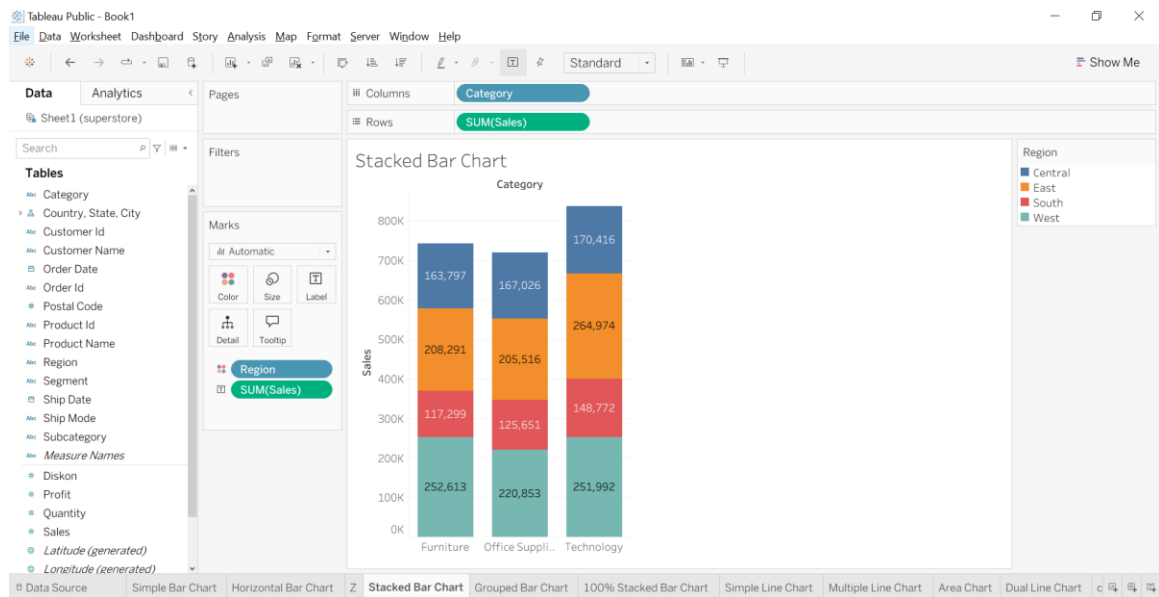
Sales → Label.



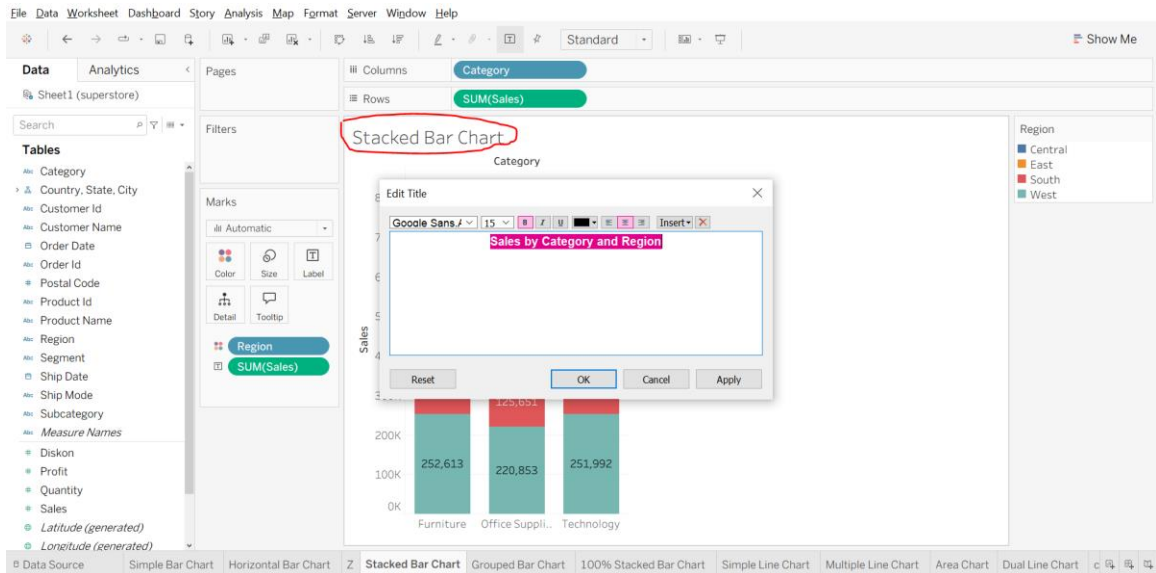
Agar Label Terbaca semua , Drag visualisasi nya ke kanan .



Sehingga Outputnya seperti ini :



Ganti title chart nya menjadi → Sales By Category and Region.



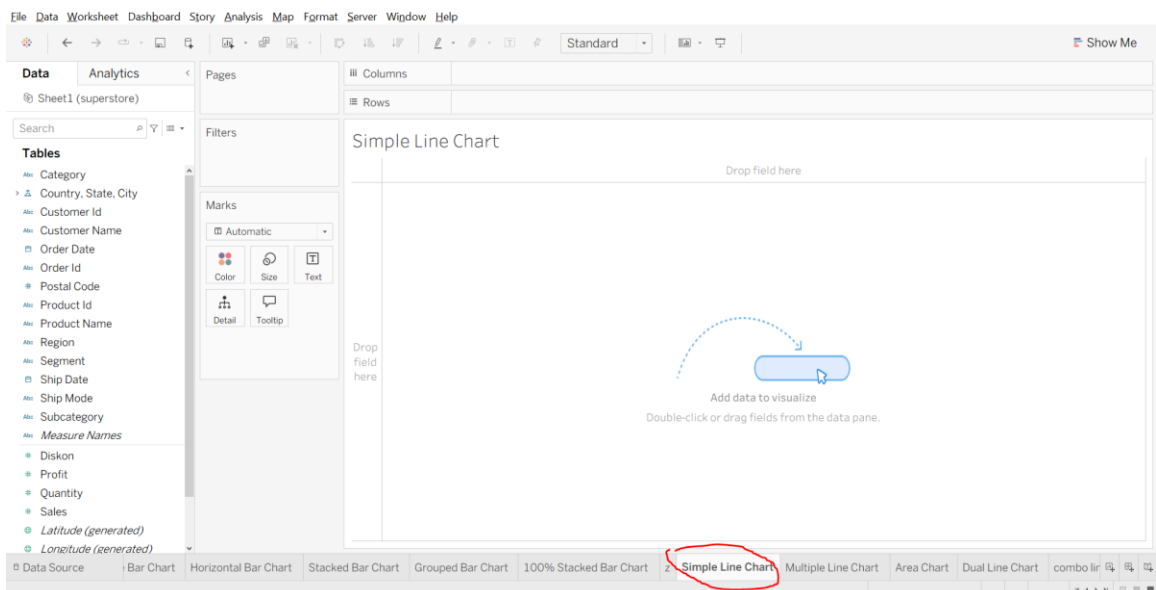
2. Line Chart

Line chart digunakan untuk melihat tren dari waktu ke waktu.

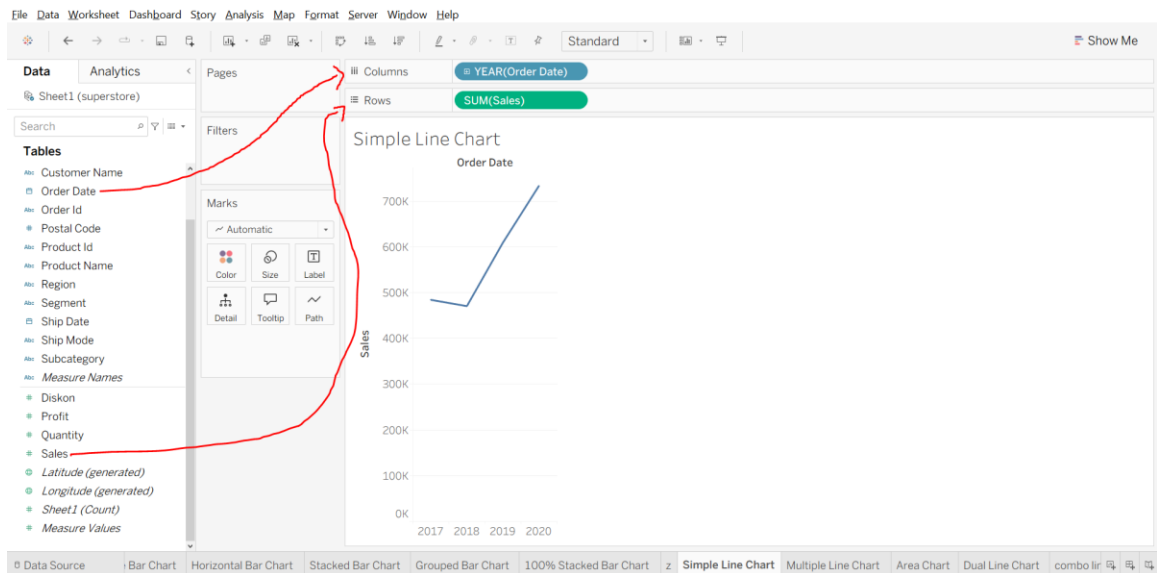
Jenis-jenis Line Chart:

- Simple Line

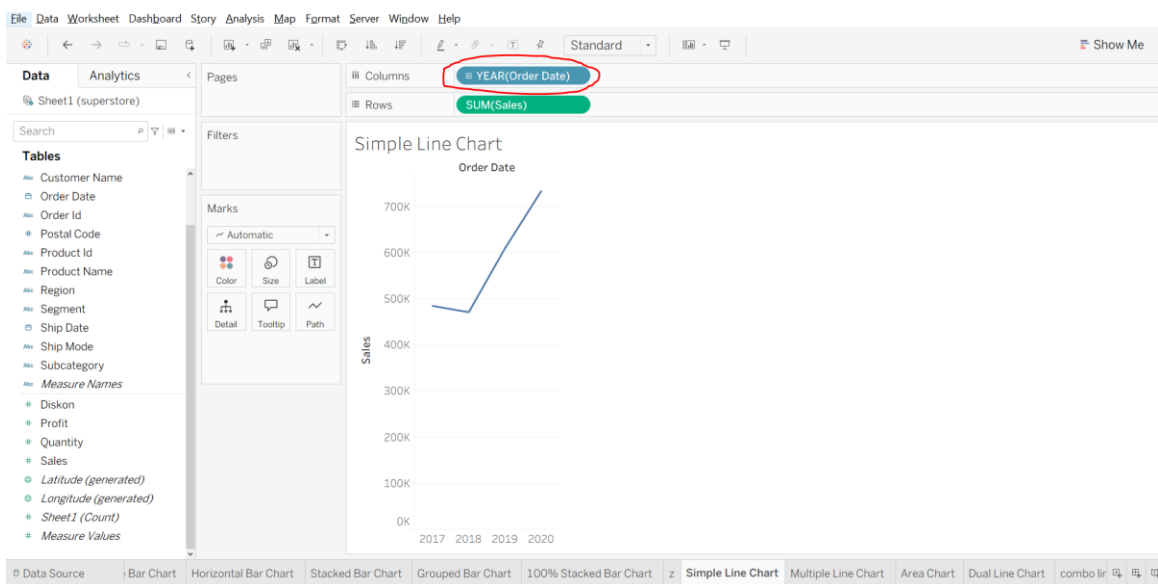
Buat sheet baru Kembali dan rename → Simple Line Chart



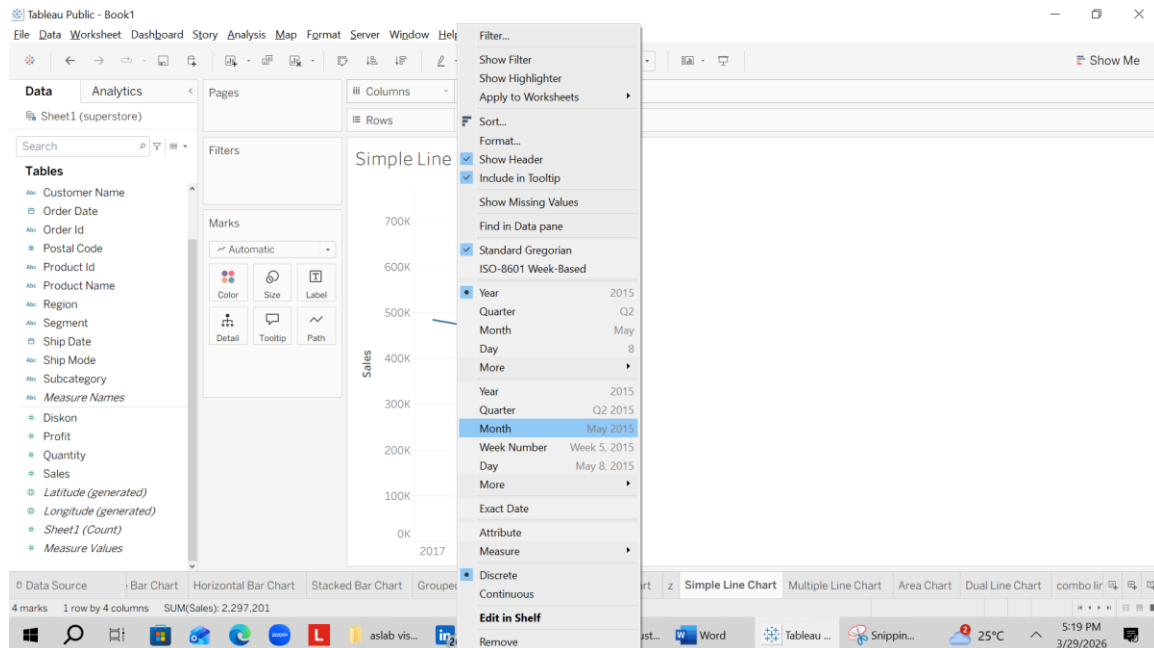
Drag & Drop Kolom:
Order Date → Columns
Sales → Rows



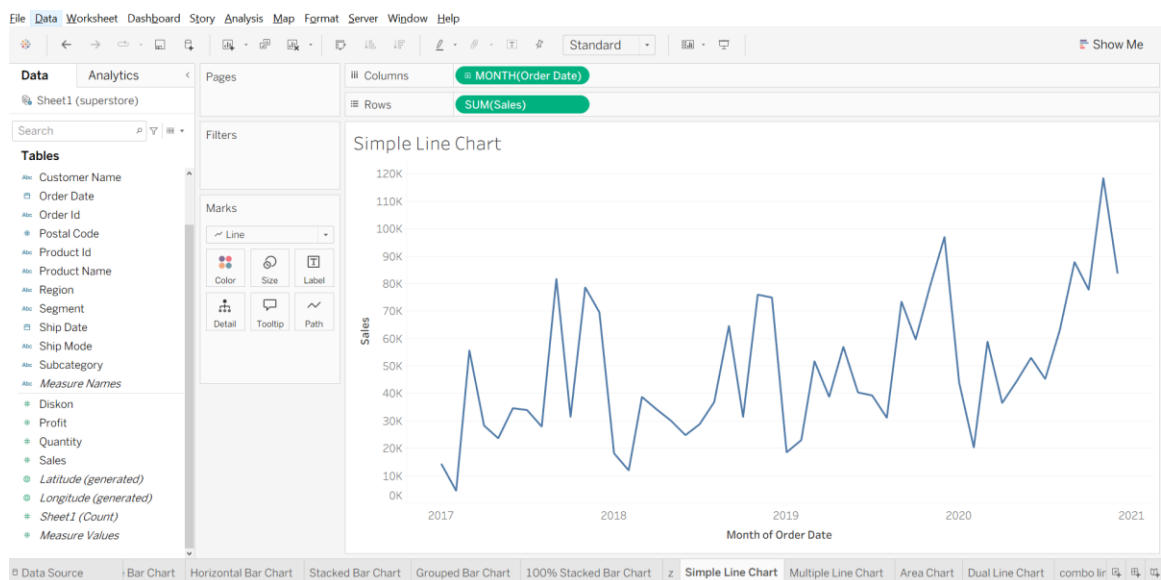
Ubah Format Date nya



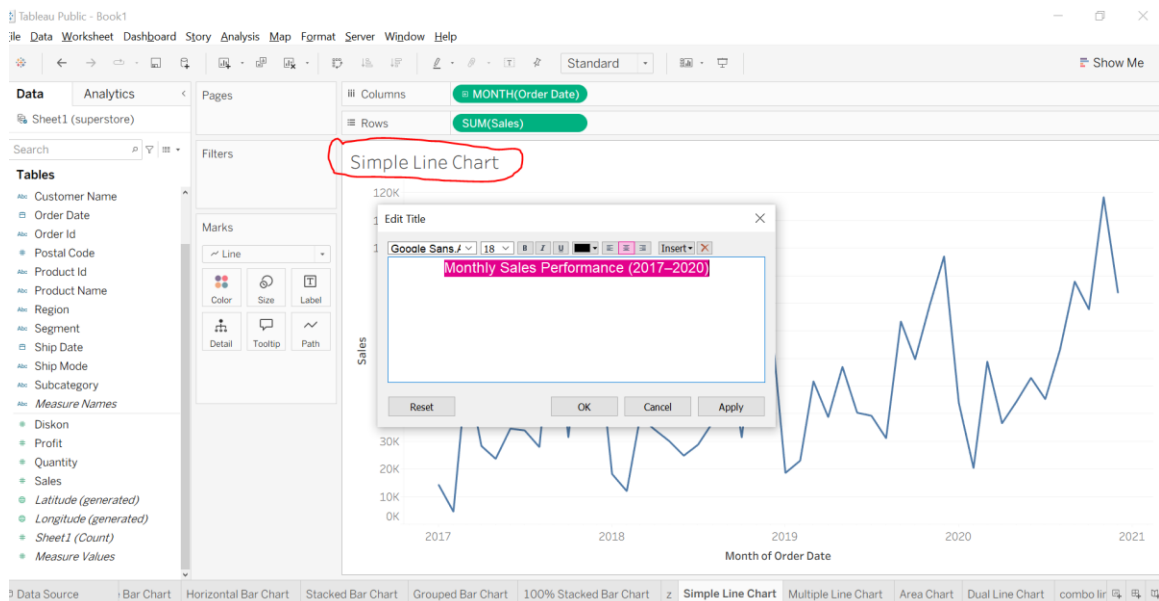
Ubah Format tanggal → Month (Continuous)



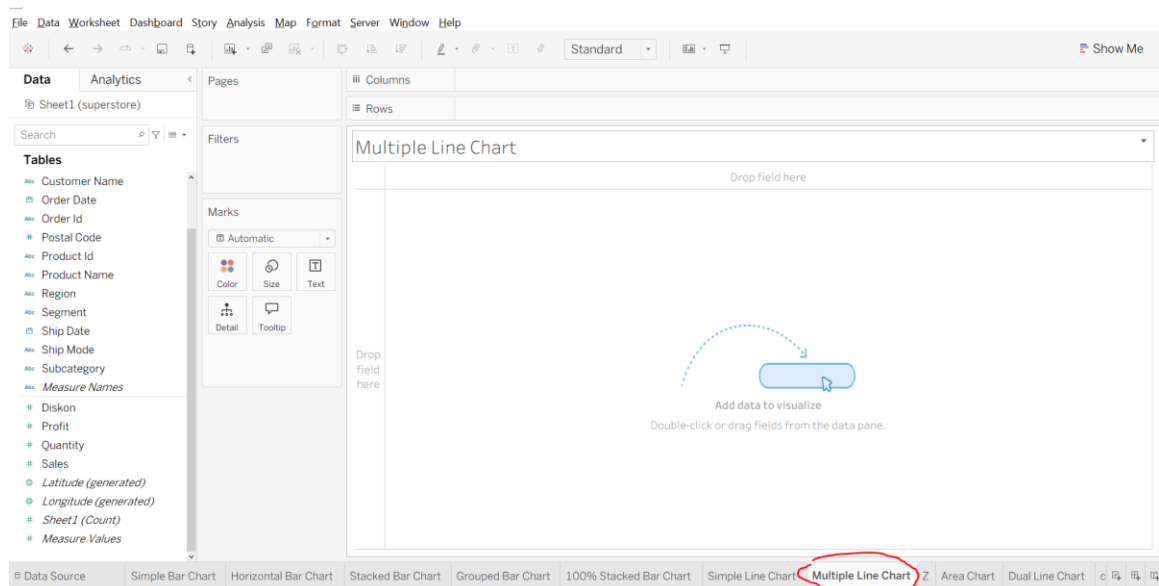
Sehingga Outputnya sebagai berikut:



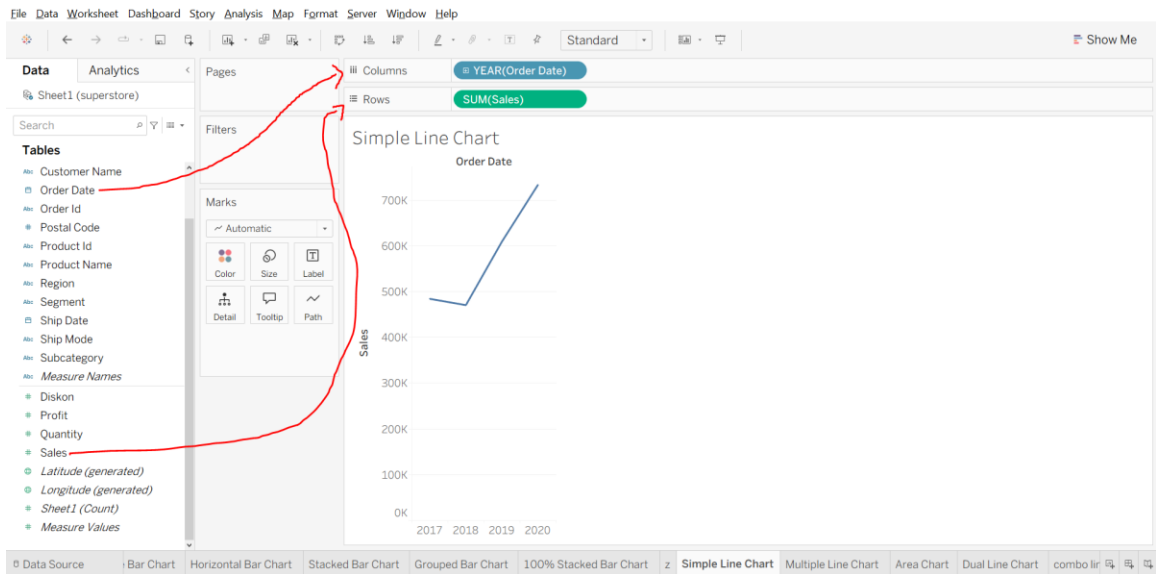
Ubah Title nya → Monthly Sales Performance (2017 – 2020)



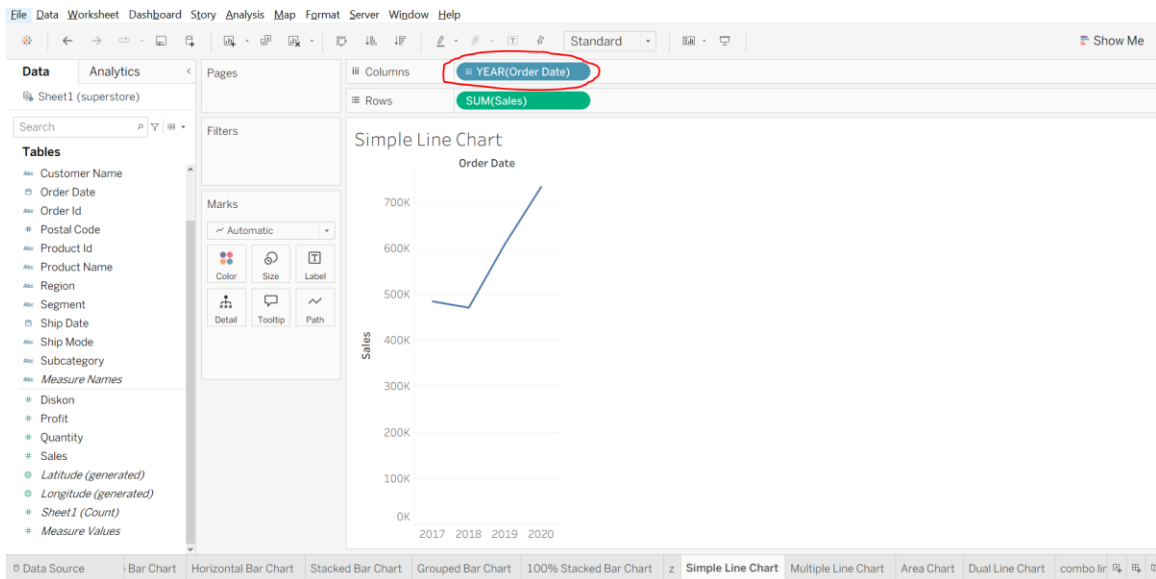
- **Multiple Line Chart**
Buat Sheet baru ganti menjadi → Multiple Line Chart

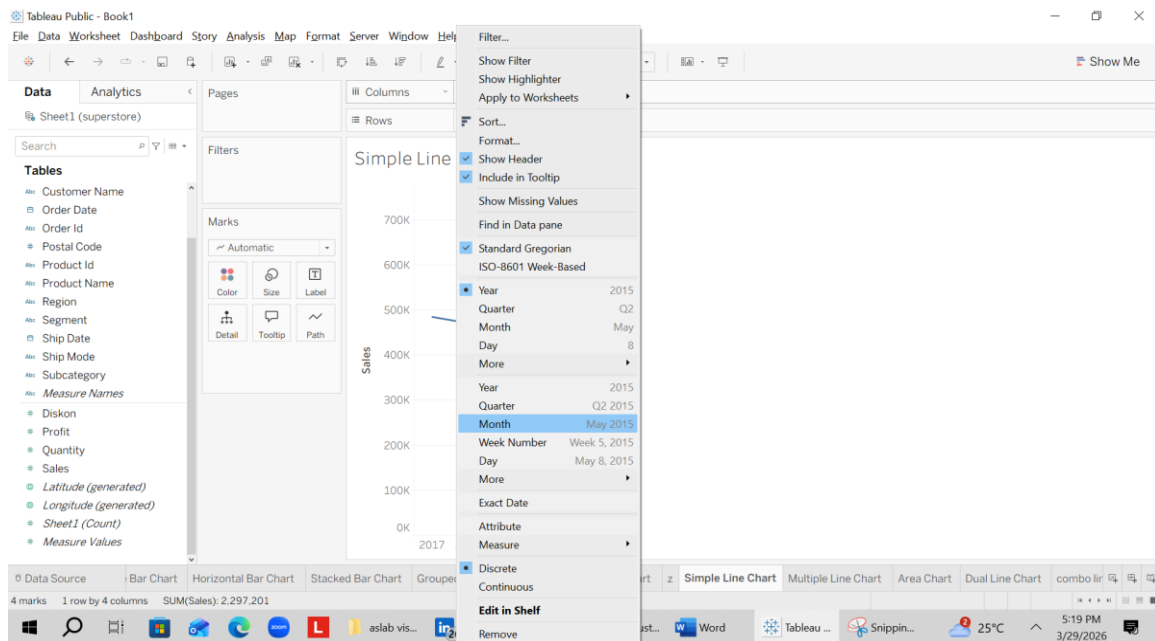


Drag & Drop Kolom:
Order Date → Columns,
Sales → Rows

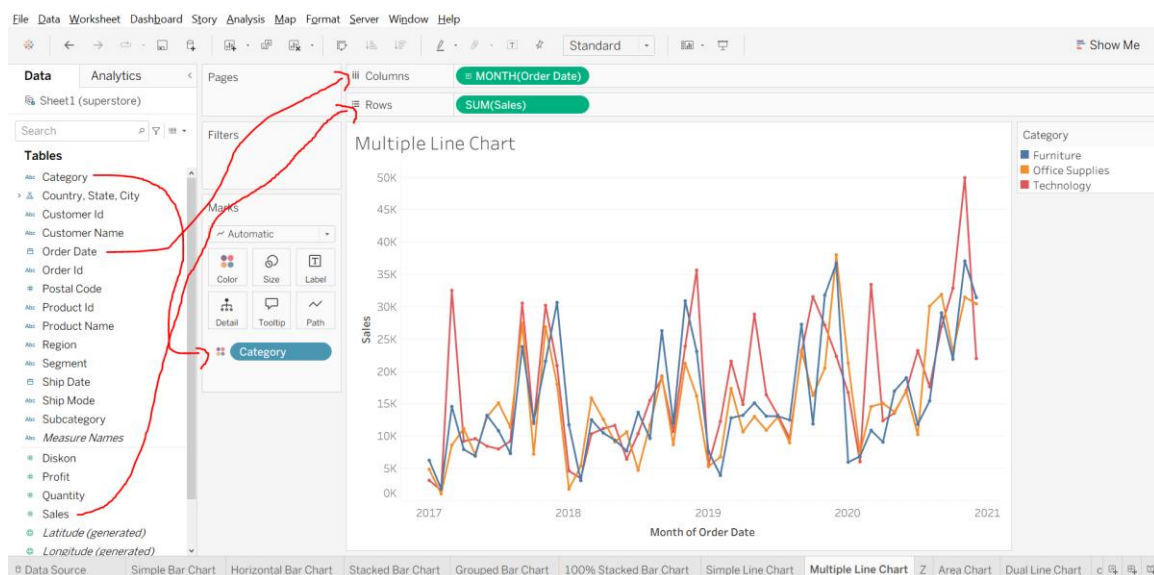


Ubah Format tanggal → Month (Continuous)

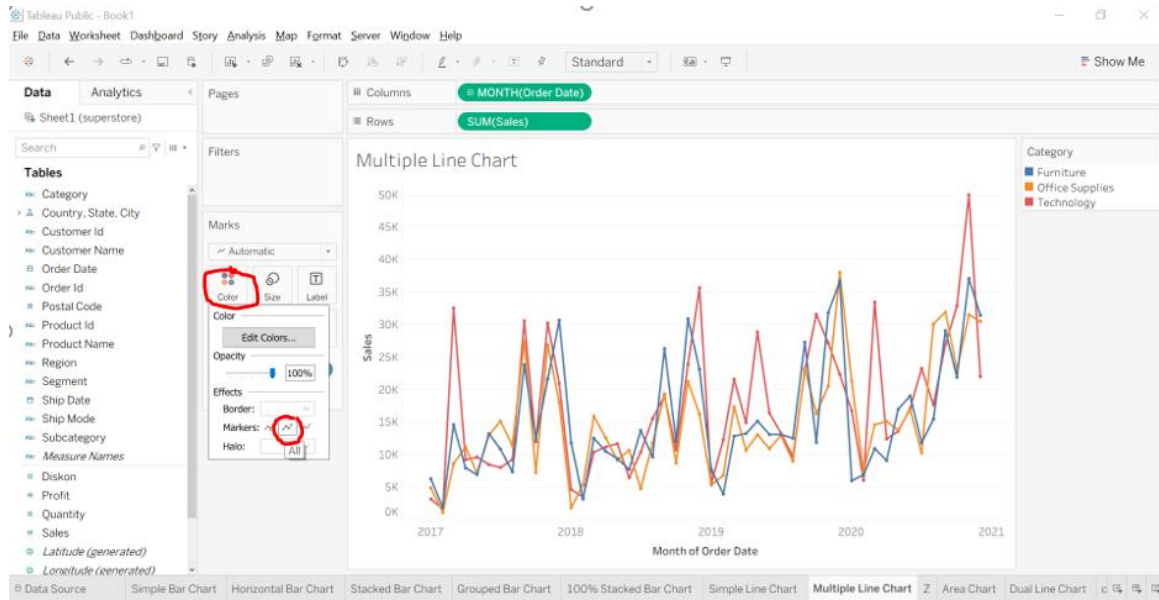




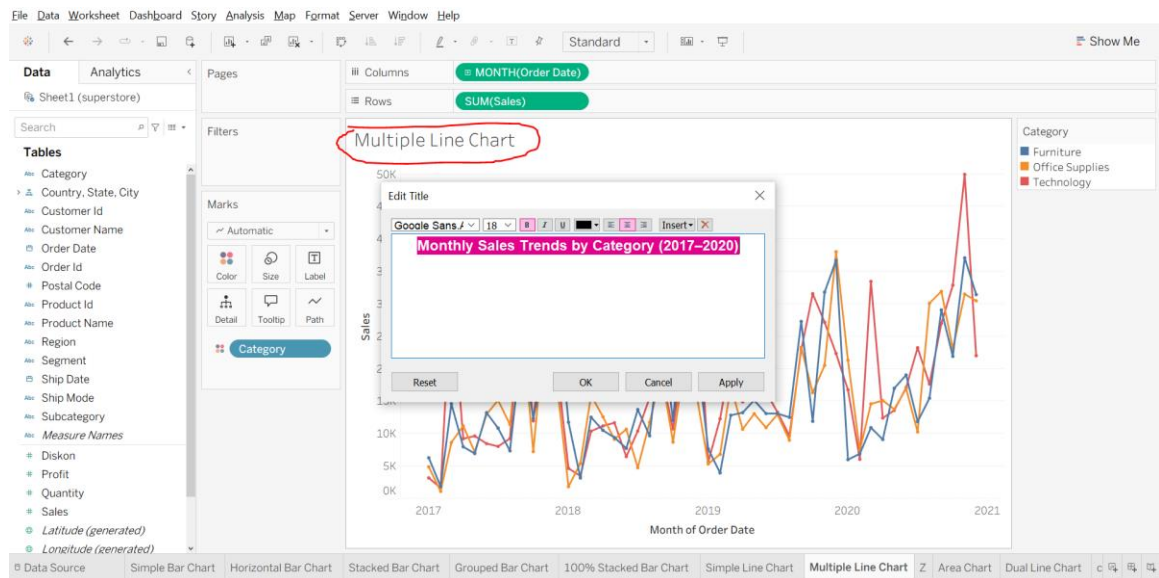
Drag & Drop Kolom:
 Category → Color (Marks),
 Order Date → Columns,
 Sales → Rows.



Ubah Markers nya sebagai berikut:

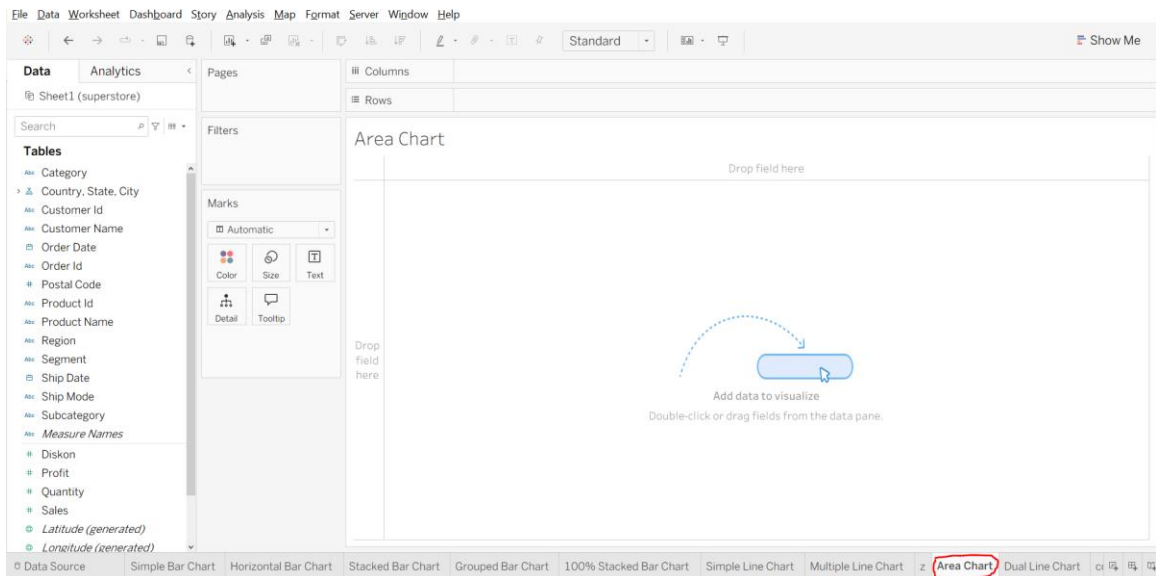


Ganti Title nya → Monthly Sales Trends By Category (2017-2020)



- Area Chart

Buat Worksheet baru, rename → Area Chart

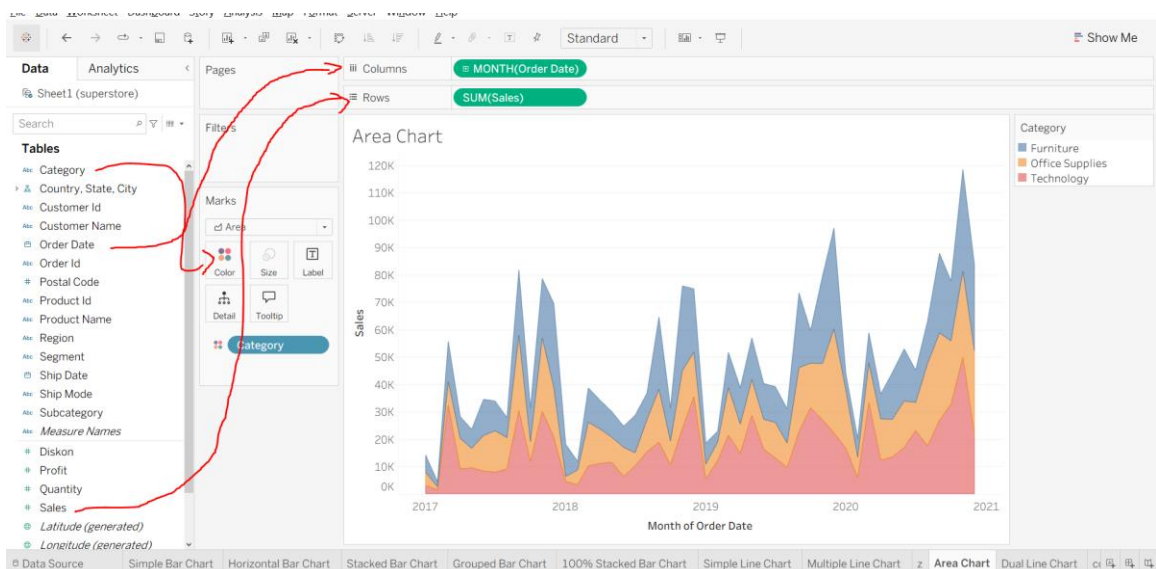


Drag & Drop kolom
Category → Color(Marks),

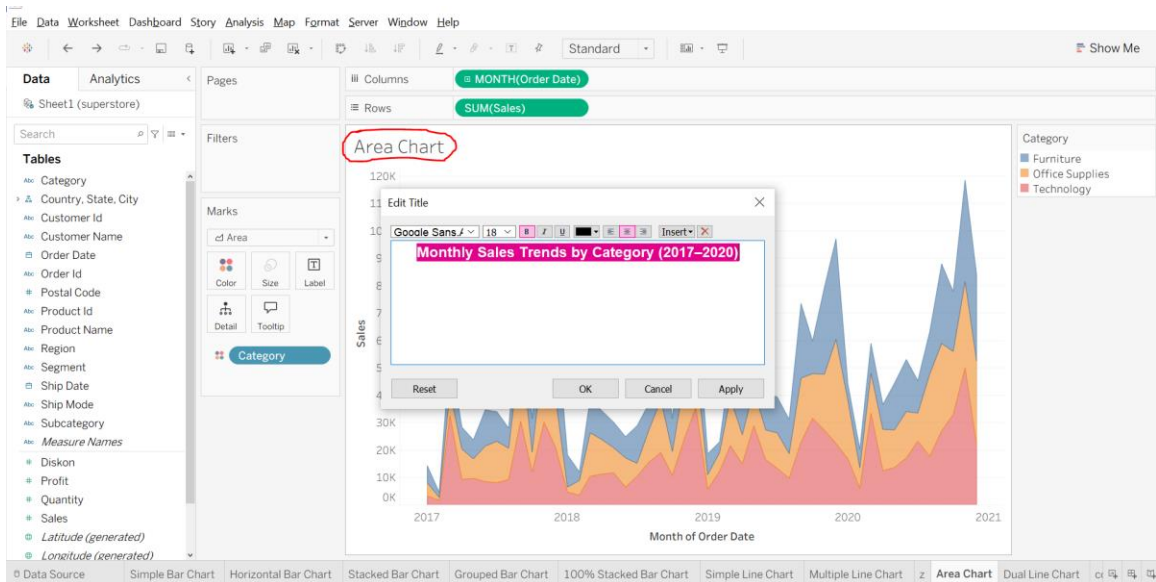
Order Date → Columns(Ubah Format tanggal → Month (Continuous),

Sales → Rows.

Pada bagian Marks → Ubah Jenis Visualisasi menjadi Area Chart

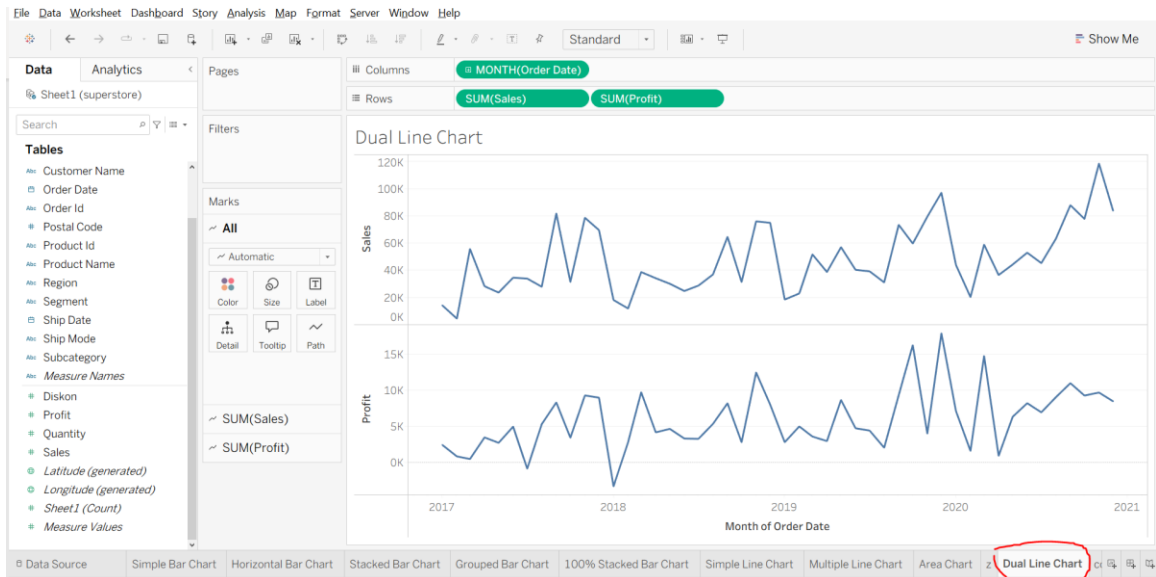


Ubah Judul Chart nya → Monthly Sales Trends By Category

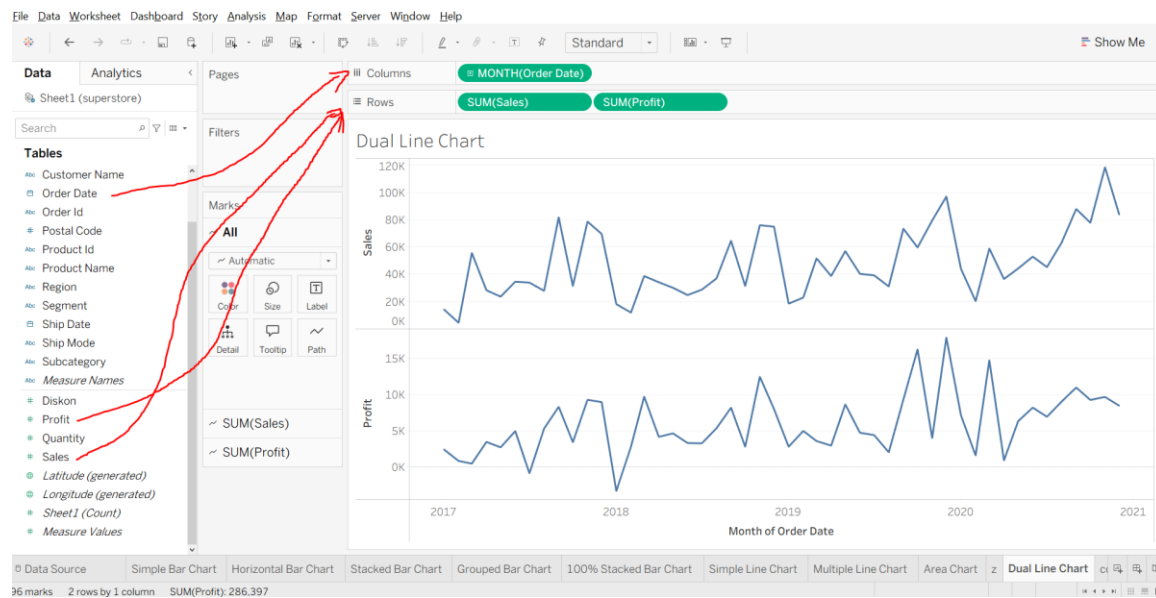


- **Dual Line Chart**

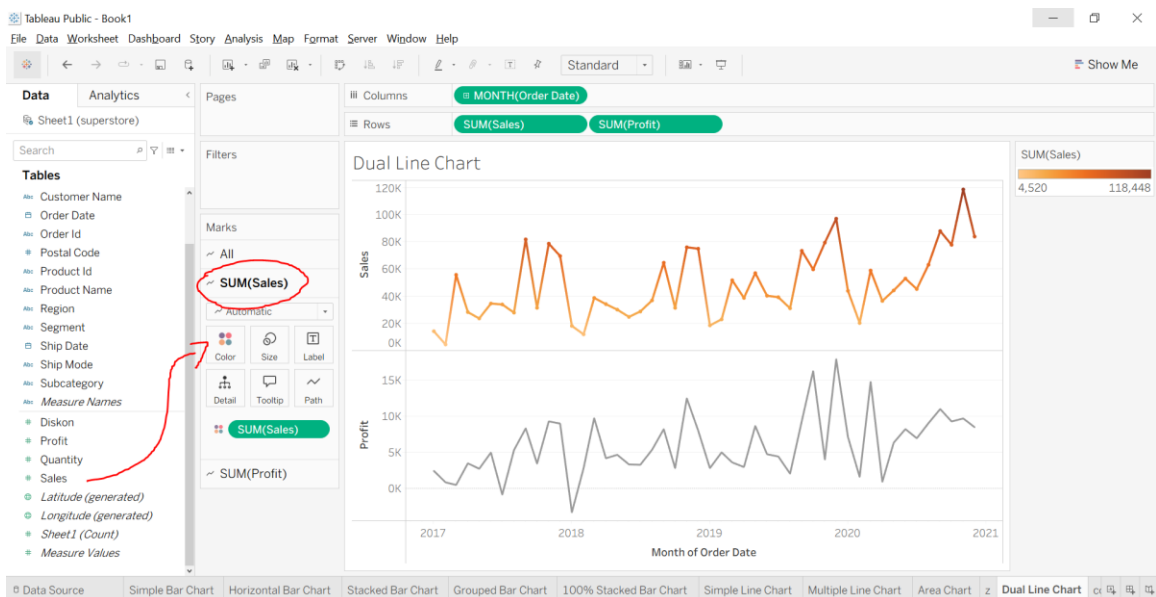
Buat worksheet baru dengan judul → Dual Line Chart



Drag & Drop kolom:
 Order Date → Columns,
 Profit → Rows,
 Sales → Rows.

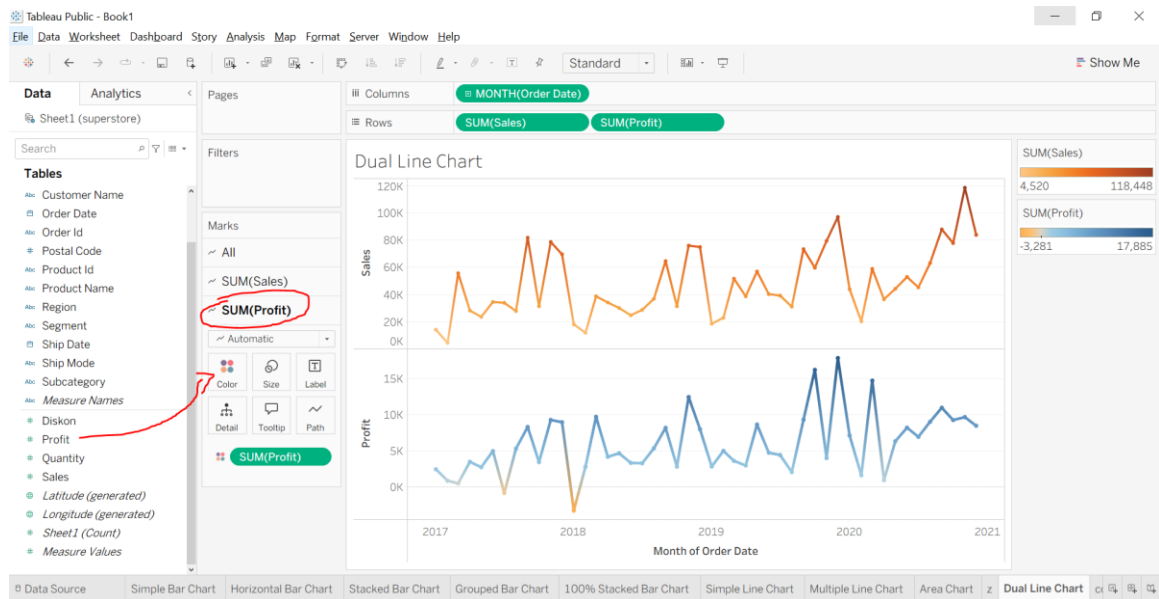


Drag & Drop Kolom:
 Sales → Marks → SUM(Sales) → Color

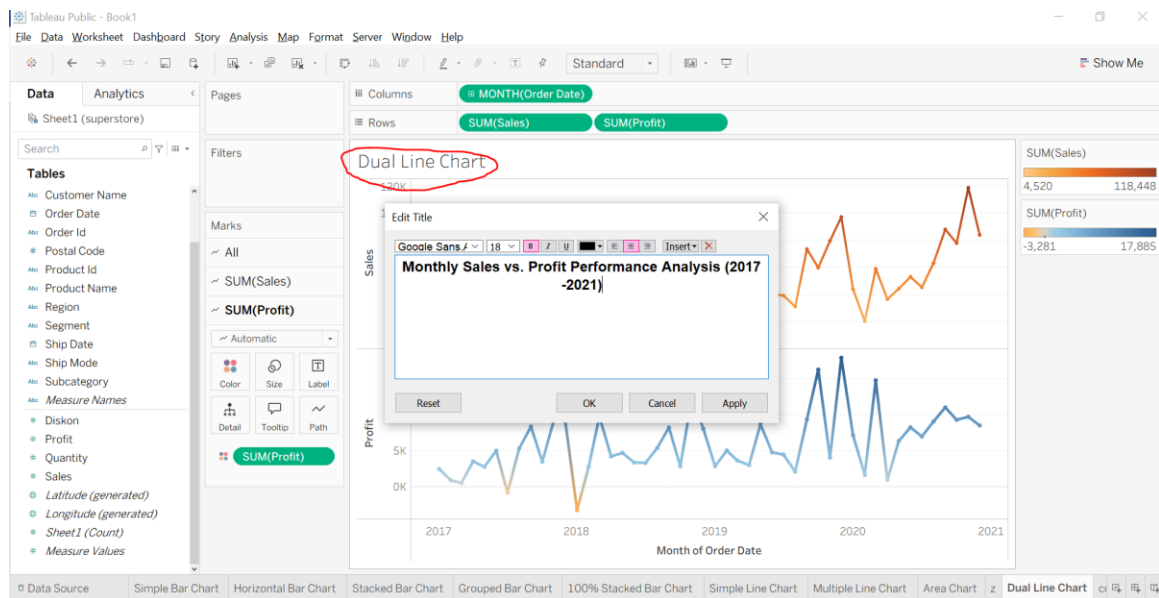


Drag & Drop kolom:

Profit → Marks → SUM(Profit) → Color

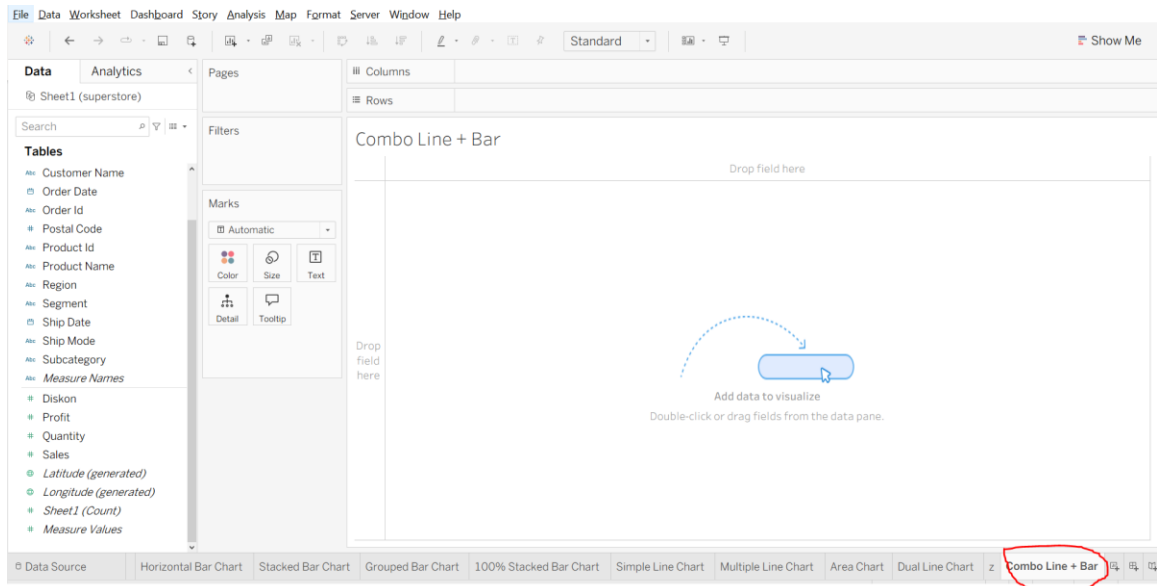


Rename Judul Chart nya → Monthly Sales vs Profit Performance Analysis (2017- 2021)



- **Combo Line + Bar**

Buat WorkSheet baru, judulnya → Combo Line + Bar Chart

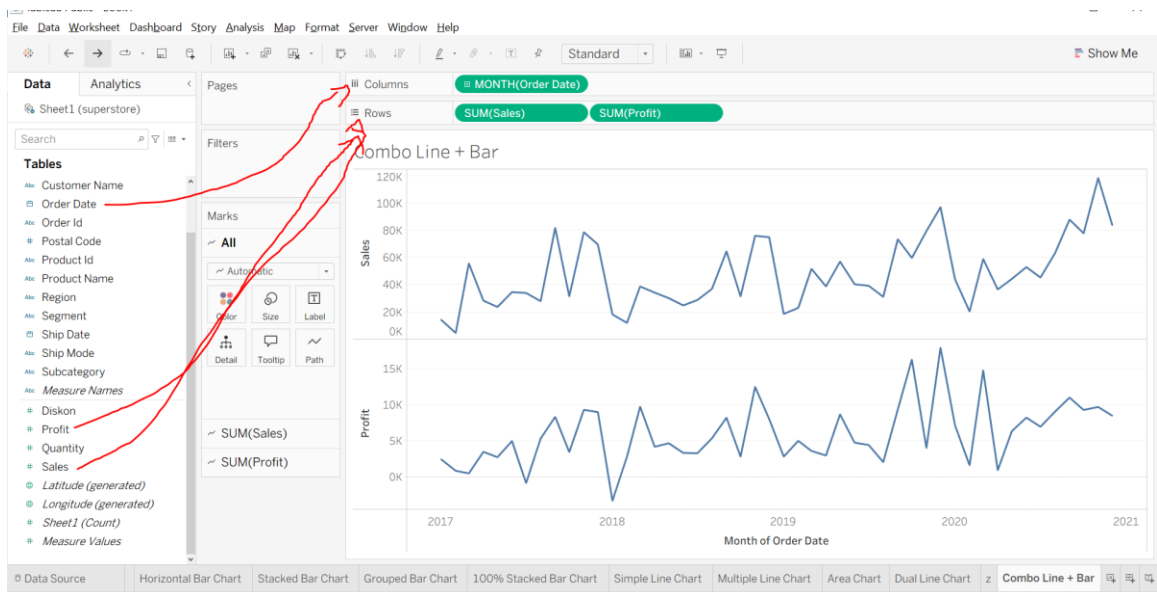


Drag & Drop Kolom:

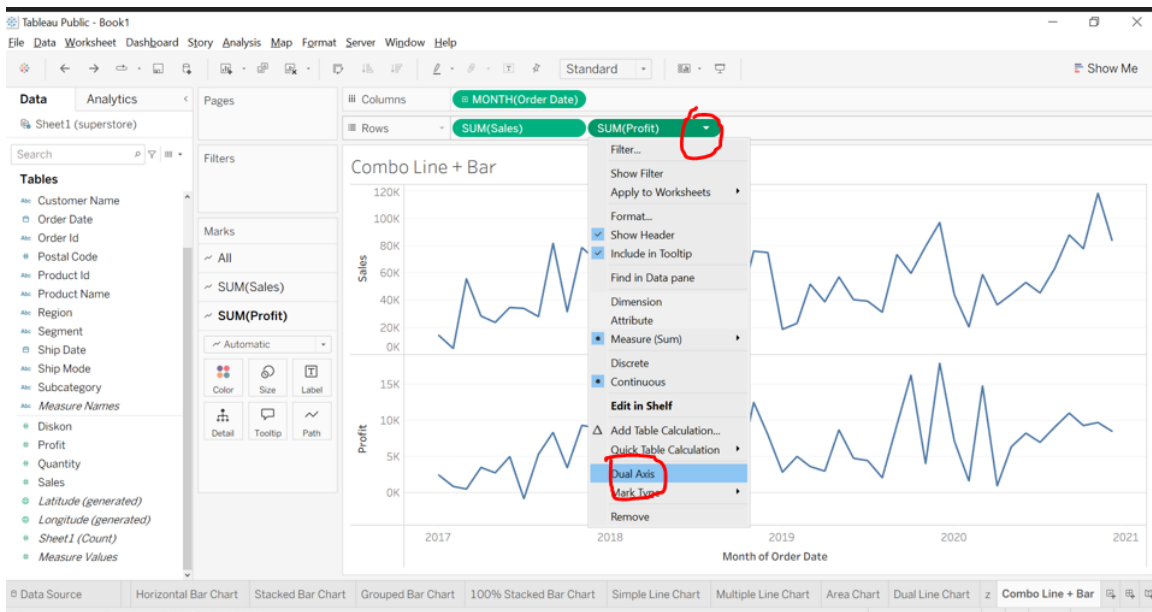
Order Date → Columns,

Profit → Rows,

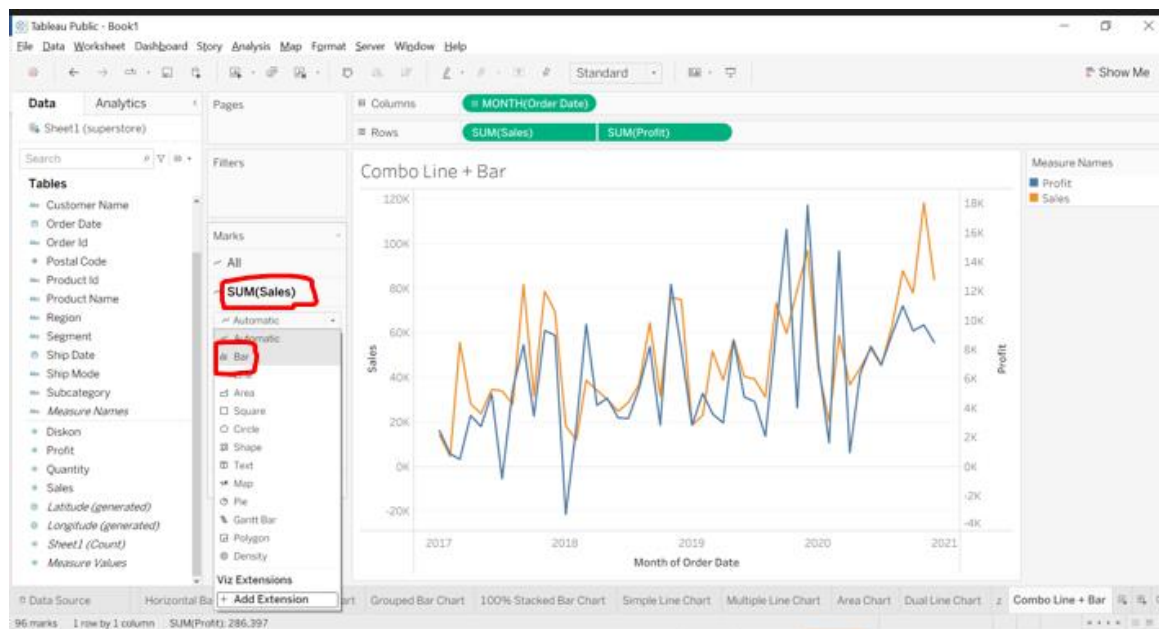
Sales → Rows.



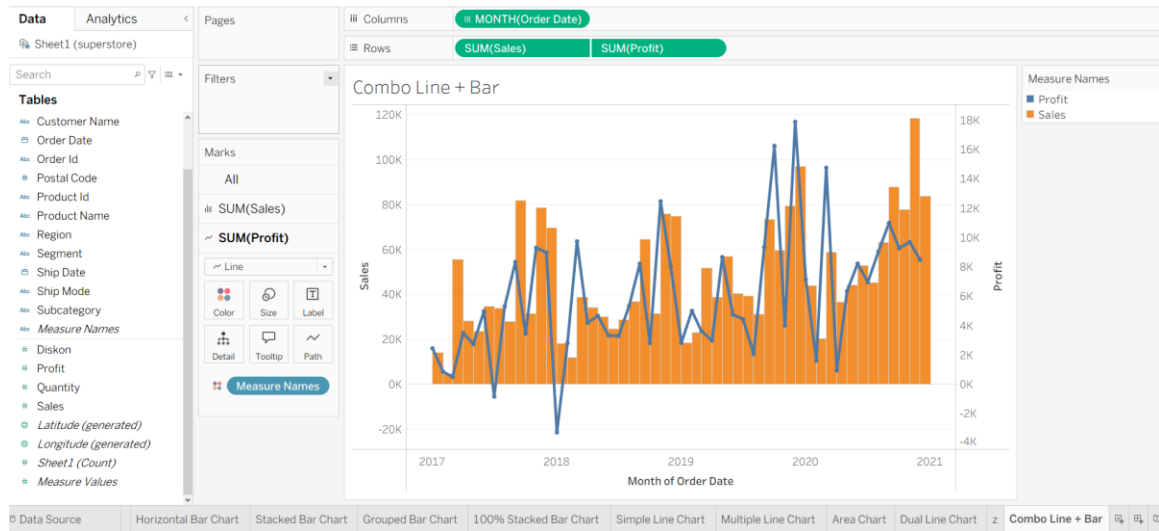
Untuk menggabungkan dua Visualisasi tersebut, gunakan dual axis. Sebagi berikut:



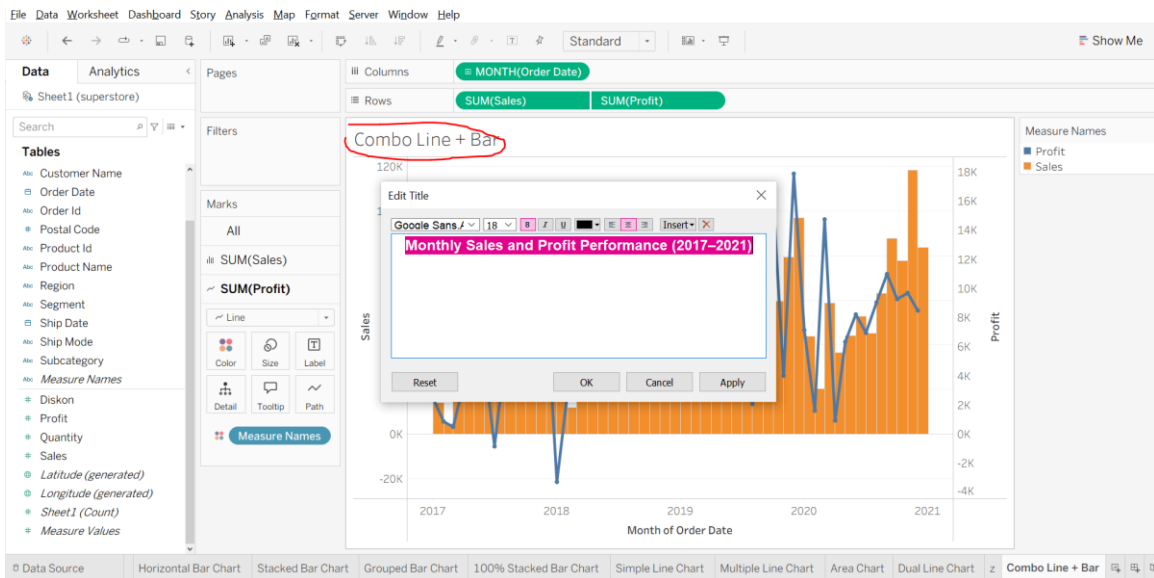
Karena Output Line Chart semua, kita ubah salah satu (Sales atau Profit) ke Bar Chart. Kita ubah sales ke Bar chart, dengan cara sebagai berikut:



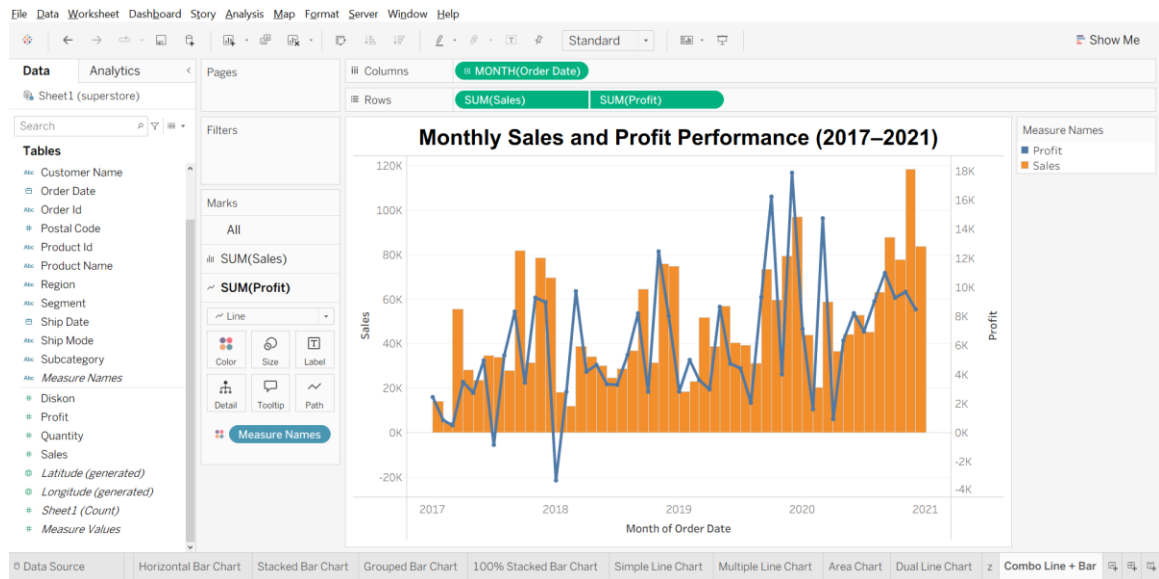
Sehingga Outputnya :



Ubah Title chartnya → Monthly Sales and Profit Performance (2017–2021)



Sehingga Outputnya :



Latihan

1. Jelaskan perbedaan fungsi dari:

- Simple Bar Chart
- Stacked Bar Chart
- Grouped Bar Chart
- 100% Stacked Bar Chart

Pertanyaan:

- a) Kapan masing-masing digunakan?
- b) Apa jenis insight yang dihasilkan?

2. Jelaskan perbedaan:

- Simple Line Chart
- Multiple Line Chart
- Line with Marker
- Area Chart
- Dual Line Chart

Pertanyaan:

- a) Kapan masing-masing digunakan?
- b) Apa kelebihan dan kekurangannya?

3. Diberikan kondisi berikut:

- Chart tidak di-sort
- Warna terlalu banyak
- Tidak ada label
- Judul tidak jelas

Pertanyaan:

- a) Apa saja kesalahan yang terjadi?
- b) Bagaimana cara memperbaikinya?
- c) Mengapa hal tersebut penting?